Shah Mohammad Rizvi

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Career Objective

Final-year BSc (CSE) student at IUBAT (CGPA: 3.82) seeking a Jr. AI/ML Developer position to apply end-to-end machine learning and deployment skills to production problems. Experienced in architecting RAG-based LLM systems, training deep-learning models for computer vision and time-series forecasting, and deploying scalable web services (FastAPI, Streamlit, MERN, Docker). Proficient in Python, TensorFlow, PyTorch, scikit-learn, LangChain, and Git; familiar with AWS/GCP/Azure. Eager to contribute to cross-functional teams to deliver measurable product impact. (Open to AI/ML internship opportunities if a more suitable entry path is recommended.)

Key Skills

Programming	Python, JavaScript, TypeScript, C++, SQL, NumPy, Pandas, scikit-learn; learning interest in R for data analysis.
Machine Learning & Deep Learning	Supervised/Unsupervised Learning, Neural Networks, Transformers, RAG Pipelines, LangChain, Vector Databases (ChromaDB), TensorFlow/Keras, PyTorch, HuggingFace, OpenCV.
Generative AI	LLM Integration, Prompt Engineering, Gemini/OpenAI APIs, Retrieval-Augmented Generation (RAG), Embedding Models, Context-Aware Chat Systems.
MLOps & Deployment	Streamlit, FastAPI/Flask, Docker, GitHub Actions, CI/CD, Render, Vercel, model serving, reproducible ML pipelines.
Full-Stack Development	MERN Stack (MongoDB, Express, React, Node), Socket.io (Real-time), Redux, Tailwind CSS, Framer Motion, JWT Authentication.
Data & Research	Data Preprocessing, Feature Engineering, Model Evaluation (Cross-validation, ROC-AUC), SHAP Explainability, Research Methodology, Technical Writing.
Tools & Platforms	Git / GitHub, Google Colab, Kaggle, Postman, MongoDB Atlas, VS Code, Azure (Fundamentals exposure).
Soft Skills	Analytical Thinking, Collaboration, Leadership (Founder of CollabCircle), Adaptability, Problem-Solving, Continuous Learning.
Languages	Bangla, English.

Selected Projects

Orbit (GenAI Research Assistant)	Designed and deployed a **RAG-powered internal assistant** for CollabCircle using LangChain and Google Gemini. Implemented privacy-first retrieval with local HuggingFace embeddings and persistent ChromaDB vector storage for querying internal policy documents. **Role:** Lead Developer **Skills:** Python, LangChain, ChromaDB, Streamlit, Docker, Git **Links:** GitHub · Live App
SolarTwin AI	Developed a **Universal Spatio-Temporal Transformer (USTT)** for multi-site solar forecasting and deployed it as a Digital Twin dashboard. Achieved cross-site scalability with <1ms inference latency using optimized TensorFlow serving. **Role:** ML Engineer **Skills:** Python, TensorFlow, Pandas, FastAPI, Streamlit **Links:** GitHub · Live App

LENSGuard	Engineered a hybrid **Network Intrusion Detection System** combining 1D-CNNs for known attack signatures and VAEs for zero-day anomalies. Built a real-time "Command Center" dashboard for continuous monitoring and threat visualization. **Role:** ML Developer **Skills:** PyTorch, scikit-learn, Streamlit, Socket.io, Docker **Links:** GitHub · Live App
SafeSkinAI	Created a computational toxicology platform predicting carcinogenicity using **XGBoost** and molecular fingerprinting. Integrated SHAP explainability for model transparency and an interactive glassmorphism-based UI. **Role:** Research Developer **Skills:** XGBoost, scikit-learn, SHAP, Streamlit, Plotly **Links:** GitHub · Live App
BidPulse	Developed a real-time auction platform supporting live bidding via Socket.io with JWT-based authentication and payment gateway integration. Showcased backend scalability and event-driven architecture. **Tech:** React, Node.js, Express, MongoDB, Socket.io Live Demo GitHub
Sex Education Discourse Dataset (US & UK)	Curated a public Kaggle dataset analyzing sex-education discourse (1922–2022) using n-gram frequency time-series. Automated extraction pipeline released under CC BY 4.0. **Tech:** Python, Pandas, Matplotlib, Kaggle API GitHub Repository

Research & Publications (selected)

- Voice-Based Parkinson's Disease Detection using SVM & XGBoost. (Paper Link)
- Reproducible Machine Learning Pipeline for Predicting Cosmetic Chemical Carcinogenicity and Reproductive Toxicity Using Molecular Descriptors and Fingerprints (Presented at 28th ICCIT 2025).
- Clinical vs. Ultrasound for PCOS Detection: A Calibrated, Leakage-Safe, Real-Time Benchmark (Presented at 28th ICCIT 2025).
- LENS-Guard: A Lightweight Hybrid VAE-CNN for High-Recall Detection of Known and Unknown Network Intrusions (Accepted at 5th ICECTE 2026)

Certifications

- Machine Learning with Python - EDGE Bangladesh (Nov 2024 - May 2025) Certificate
- Kaggle Courses: Feature Engineering, Data Visualization; Computer Vision; Intro to Deep Learning; Intermediate ML; Intro to Machine Learning; Pandas; Python (2024-2025) Certificate
- Artificial Intelligence Fundamentals (IBM SkillBuild, 2025) Certificate
- Machine Learning Using Python (Simplilearn, 2025) Certificate
- Machine Learning for Beginners (Simplilearn, 2025) Certificate

Training & Workshops

- Machine Learning with Python - EDGE Bangladesh (Nov 2024 - May 2025) Certificate
- Seminar: Charting the Azure Horizon - Styava.dev & Microsoft (10 Dec 2024) Certificate
- Seminar: Generative AI in Copilot Studio - Styava.dev & Microsoft (7 Nov 2024) Certificate

Activities

- Participant - Solvio AI Hackathon (2025)
- Participant - ICPC Dhaka Regional Preliminary Contest (2024)
- Volunteer - OBHIZATRIK Foundation (2022-2023)
- Cultural Captain - Karnafuli House, BCIC College (2018-2019)

Education

2022-present	BSc in Computer Science & Engineering, **IUBAT**	(CGPA: 3.82 current)
2020	Higher Secondary Certificate, **BCIC College**	(GPA: 5.00)
2018	Secondary School Certificate, Pallabi Bidya Niketan	(GPA: 5.00)

Hobbies

- Watching Movies
- Exploring AI research updates
- Seeing Friends

References

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